CONTACT

Email
makalicardoso@gmail.com

Phone
+27 68 262 2669

Location
Cape Town, South Africa

Portfolio
plandimakali.netlify.app

LinkedIn
linkedin.com/in/plandimakali

GitHub
github.com/Dimak12

TECHNICAL SKILLS

AI & Machine Learning
Machine learning, deep learning (MLP, CNN, LSTM), graph neural networks, natural language processing, exploratory data analysis, PyTorch, TensorFlow / Keras

Software Development
Python, Java, JavaScript, Vue.js, C / C++, PHP, HTML / CSS, MySQL, REST APIs, Microsoft Azure

Networking & Systems
Software-defined networking, OpenDaylight, Cisco IOS XRv, network protocols, CCNA 1 & 2, intrusion detection

Tools & Productivity
Git / GitHub, Ubuntu, MATLAB, Excel, Word, PowerPoint, Power BI (introductory)

LANGUAGES

English	Fluent
French	Native / bilingual



Plandi Makali

MSc Artificial Intelligence Student · Computer Engineer

"Solving Today's Challenges, Shaping Tomorrow's Solutions"

Profile

Computer engineer and MSc Artificial Intelligence student at the University of Cape Town, interested in machine learning, deep learning and explainable AI. Background in software development and networking, with industry experience in backend and intelligent systems development. I want to build intelligent systems that solve real-world problems.

Education

MSc in Artificial Intelligence (Coursework and Dissertation)
University of Cape Town — 2026 – Present

Distinctions in all 1st semester modules. Dissertation in progress on logic-based explanations for neural network predictions.

BEng Tech Honours in Computer Engineering ↗
Cape Peninsula University of Technology — 2024 – 2025

Average: 80%. Thesis: deep learning-based intrusion detection for hybrid software-defined networks.

Bachelor of Engineering Technology in Computer Engineering ↗
Cape Peninsula University of Technology — 2022 – 2023

Cum Laude. Average: 82%.

Diploma in Engineering Technology in Computer Engineering ↗
Cape Peninsula University of Technology — 2020 – 2021

Average: 85%.

Academic Achievements

- **Dean's Excellence Award (2023)** ↗**:** Best BEng Tech student, Department of Electrical, Electronic and Computer Engineering
- **Dean's Excellence Award (2021)** ↗**:** Top achiever, Diploma in Computer Engineering
- **33 distinctions from 38 courses** across the Diploma, BEng Tech, Honours and MSc
- Consistently ranked in the top 1% of the class (Diploma and BEng Tech)

VOLUNTEERING

IEEE GLOBECOM 2024
Volunteer · Cape Town
Front desk operations, logistics and session monitoring; supported speakers and presenters.

AfricArena 2023 Grand Summit
Coordinator · Cape Town
Front desk operations and logistics.

Melkbosstrand Beach Clean-up
Volunteer · Melkbosstrand

CERTIFICATES

African Management Institute
- Problem Solving
- Communication
- Personal Productivity

Microsoft Office Associate

SOFT SKILLS

- Communication and collaboration
- Problem solving and analytical thinking
- Leadership and productivity
- Detail-oriented, fast learner

INTERESTS

Artificial intelligence, reading, running, tech and innovation blogs

REFERENCES

Adrian Wyngaard
Lecturer, Cape Peninsula University of Technology
WYNGAARDA@cput.ac.za

Ayodele Periola
Lecturer, Cape Peninsula University of Technology
periolaa@cput.ac.za

Plandi Makali

makalicardoso@gmail.com

Professional Experience

Backend & Intelligent Systems Development
Koolboks Nigeria — Jan 2025 – Feb 2026

- Supported the development of intelligent system solutions and product features.
- Applied AI-driven solutions in real-world contexts.
- Led projects, strengthening project management and strategic thinking.
- Gained hands-on experience in system integration and digital infrastructure.

Projects

Deep Learning IDS for Hybrid SDN — Honours thesis
CNN-based intrusion detection integrated with an OpenDaylight controller. 99.98% detection accuracy, with false alarms cut from 13,643 to 5.

Handwritten Digit Recognition (ANN)
Neural network trained on MNIST (about 97.6% test accuracy) and tested on custom handwritten digits.

School Bus Monitoring System
IoT system using RFID, ESP32, GPS and an Azure-hosted database, with SMS alerts to parents.

Goodies Store
Vue.js e-commerce web app with a REST API product catalogue and a shopping cart.

Solar Charge Controller
Arduino-based PWM controller that charges a 12V battery with overcharge protection.

ServiceMe
Web platform for booking house cleaning and gardening services (Java, MySQL).

Connect
Twitter-like social platform built with JSP, Servlets, MySQL and AJAX.